**Exercise Set 4** 6 October 2025

# 1 Weakest Precondition (with loops!)

## 1.1 Computing WP (with loops)

Given the following functions

```
stupidadd(a,b) =
  while (b > 0) do
    a := a + 1;
    b := b - 1
  done;
  a
```

```
eucliddiv(a,b) =
  quot := 0;
  rest := a;
  while (rest >= b) do
    rest := rest - b;
    quot := quot + 1
  done;
  q
```

- Give some valid invariants for the two given functions
- What about instead variants?
- Code the function in why3 and add the obtained invariants/variants

## 1.2 A sorting algorithm in why3

Now that we have all the tools to manage pieces of code with loops, you are ready to implement the bubblesort!
Here a pseudocode of the bubblesort algorithm

```
bubblesort(a) =
  i := a.length-1;
  while (i >= 1) do
    j := 0;
    while (j <= i-1) do
      if (a[j+1] < a[j]) then
        aux := a[j+1];
        a[j+1] := a[j];
        a[j] := aux
      endif;
```

```
    done;
  done;
  a
```

Proceed step by step with the implementation on why3

- Implement it on why3
- What should be the postconditions? How the input array and the output sorted array are related?
- We have two nested while loops, hence we need to define invariants for every while loop

Now you can go crazy and implement also Insertion sort!